

A Lightweight Dual-Branch Multi-Level Attentive Attributes With Constraint Fusion Network for EEG-Based Alzheimer's Detection

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Abstract—Alzheimer's disease (AD) is a degenerative neurological illness that causes cerebral atrophy, neuronal death, and progressive cognitive decline, making it the predominant cause of dementia. Accurate classification of electroencephalogram (EEG) signals is crucial for the early detection of AD and its intermediate stage, Mild Cognitive Impairment (MCI) (or Frontotemporal Dementia, FTD). This study introduces LDANet, a lightweight deep learning model designed for multiclass classification of EEG signals into Normal, MCI (or FTD), and AD categories utilizing three publicly available datasets. EEG recordings are segmented into 4-second windows and reshaped into 2D matrices representing channel-wise temporal information. LDANet employs low-level (LL) and high-level (HL) feature extraction through convolutional layers and a residual-dilated spatial pyramid pooling module to capture discriminative characteristics. Spatial and channel-wise attention mechanisms refine HL and LL features, and a constraint-based fusion adaptively combines them for final classification. The model is evaluated under three experimental setups: subject-dependent, subject-independent, and cross-dataset protocols. Results demonstrate strong performance across all configurations, and outperformed state-of-the-art approaches, as well. The findings suggest robustness, efficiency, and suitability for practical early AD diagnosis.

Note to Practitioners—Early detection of AD and its intermediate stage, MCI (or FTD) is vital for timely intervention. EEG-based diagnostic systems offer a non-invasive and cost-effective solution but often suffer from limited generalizability and high computational demands. This paper presents LDANet, a lightweight deep learning model designed for efficient multiclass classification of EEG signals into Normal, MCI (or FTD), and AD. LDANet extracts rich spatial-temporal features using residual-dilated spatial pyramid pooling and dual attention mechanisms, without converting EEG into image formats. A constraint-based fusion strategy further enhances feature representation while maintaining low complexity. Evaluated across subject-dependent, subject-independent, and cross-dataset protocols, LDANet consistently outperforms existing methods, demonstrating robustness and practical applicability for real-world clinical deployment.

Index Terms—Alzheimer's disease, mild cognitive impairment, electroencephalogram, multiclass classification, attentive features, constraint fusion.

I. INTRODUCTION

DEMENTIA represents a group of progressive neurodegenerative disorders that impair memory, thinking, and behavior, posing a growing global health challenge. Among its various forms, such as Alzheimer's Disease (AD), Mild Cognitive Impairment (MCI), and Frontotemporal Dementia (FTD), AD is the most prevalent and severe type. It is a slowly progressing condition that primarily affects memory and cognitive function, leading to loss of independence and severely reduced quality of life in older adults. In 2021, AD ranked as the seventh leading cause of death worldwide, underscoring its substantial societal and economic burden. Within the global dementia landscape, India held the fourth-largest share in 2019 and, due to its rapidly aging population, is projected to become the second most affected country by 2050; surpassing both the United States and Japan [1]. These alarming statistics highlight the urgent need for early and accurate diagnostic solutions to slow disease progression and enable timely intervention [2]. Despite significant progress in neuroscience, early and reliable diagnosis of AD remains a major challenge. Clinical evaluations primarily rely on cognitive assessments and patient history, while definitive confirmation often occurs after postmortem through detection of amyloid plaques and tau tangles [3]. Although biomarker-based methods such as cerebrospinal fluid analysis, magnetic resonance imaging, and positron emission tomography have improved diagnostic precision; their high cost, invasiveness, and infrastructural requirements limit routine use, especially in low-resource clinical environments [4]. However, these methods are not well-suited for large-scale, early-stage screening or continuous monitoring, creating a significant gap in accessible, noninvasive, and real-time diagnostic solutions for AD.

These limitations have led to growing interest in affordable, noninvasive, and scalable alternatives for early-stage detection. Among emerging modalities, electroencephalography (EEG) offers a promising approach for identifying neurophysiological alterations linked to AD [5], [6], [7], [8]. EEG is cost-effective, portable, and capable of real-time brain monitoring with high temporal resolution. It captures subtle abnormalities in neuronal oscillations and connectivity that reflect early cognitive

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decline, making it particularly suited for large-scale screening and continuous monitoring of neurodegenerative progression. Despite notable progress in deep learning for EEG-based AD detection [9], several limitations remain unaddressed. Existing EEG-based approaches often require long-duration recordings, complex preprocessing, or heavy models that limit their practicality and generalization across datasets. Previous studies mostly rely on single-channel EEG signals, trained on small-scale datasets with limited participant diversity, and often rely on complex architectures, making it difficult to generalize findings across broader clinical populations. Most existing works focus solely on binary classification, typically distinguishing between Alzheimer's and healthy controls, while research on multiclass classification, which includes intermediate stages like MCI, remains relatively scarce. This indicates a clear research gap for lightweight, interpretable frameworks capable of learning discriminative EEG patterns from short-duration signals. Motivated by this gap, the present study introduces a Lightweight Dual-Attention Network (LDANet) that leverages both low-level (LL) and high-level (HL) EEG representations for robust AD identification using short 4-second recordings, offering a practical pathway toward scalable, early, and accessible diagnosis. The proposed model is trained and tested on three different public datasets under three evaluation techniques i.e. subject-dependent, subject-independent, and cross-dataset, to verify robustness across varied clinical scenarios while keeping the parameter count well below 2.6 million and inference times suitable for real-time use.

The primary contributions of this work are as follows:

- 1) A novel LDANet architecture is proposed for Alzheimer's detection using only 4-second EEG recordings, making early screening practical and deployment-ready.
- 2) LDANet uniquely combines LL and HL EEG representations within a unified structure, enabling simultaneous capture of fine-grained temporal cues and broader contextual patterns that are often overlooked in existing models. Further, an RDSPP module is introduced to enhance LL and HL feature representation for enabling richer temporal-spectral characterization.
- 3) Channel and spatial attention modules are integrated to emphasize the most informative EEG patterns and suppress irrelevant responses, highlighting the most relevant and distinctive features.
- 4) A constraint-driven adaptive fusion mechanism is utilized to optimally balance contributions from complementary feature levels.
- 5) Evaluations across subject-dependent, subject-independent, and cross-dataset protocols show consistent gains over state-of-the-art (SOTA) methods with only 2.19M parameters, while t-SNE visualization highlights clear separability among cognitive states.

We organize the subsequent sections of this article as follows. Section II discusses SOTA approaches. The proposed LDANet framework is described in Section III. The details of the experiment results, which include the comparison-analysis and ablation tests, are presented in Section IV. This work is concluded in Section V.

II. RELATED WORK

Efforts to accurately classify AD from EEG signals have grown significantly in recent years. Table I summarizes key studies on detecting AD using EEG data. In [10], Casani and Falk divided the EEG spectrograms into localized time–frequency patches to extract spectral–temporal modulation characteristics reflective of AD severity and fed into a support vector machine (SVM) for classification. Li et al. [11] combined variational autoencoder (VAE) with fuzzy classifier to perform binary classification. However, it depends heavily on the precise tuning of VAE hyper-parameters. Cejnek et al. [13] employed a one-class SVM method to identify AD, MCI, and cognitively normal (CN) class, achieving an accuracy of 90.67%. In [14], Rodrigues et al. utilized the Lacsogram tool that analyzes lacunarity features and artificial neural network is applied on extracted features achieving an accuracy of 90%. Klepl et al. [15] presented a Graph Neural Network (GNN) for AD classification. They systematically compared six functional connectivity methods (including coherence, correlation, and PLV) for constructing EEG-based brain graphs achieving an accuracy of 92%. Araújo et al. [16] created a data-driven system for AD classification employing resting-state EEG recordings, wavelet packet decomposition, and nonlinear features. The use of classical machine learning (ML) models and CNN led to 93.8% accuracy in binary classification. However, a small sample size ($N=38$) and lower multi-class accuracy (56.8%) limited the practical application despite great performance. Dogan et al. [17] considered a graph-inspired primate brain pattern (PBP) feature extractor combined with tunable Q-factor wavelet transform (TQWT) for multilevel feature generation. The model achieved 92.01% accuracy with a weighted k-NN classifier. Lopes et al. [18] suggest a CNN-based model for identifying AD biomarkers using saliency maps and EEG power modulation spectrograms achieving an accuracy and F1-score of 93.0% and 91.6% respectively. Klepl et al. [19] presented an Adaptive Gated Graph Convolutional Network (AGGCN) for explainable diagnosis of AD using EEG data. The model learns brain connectivity graphs from power spectral density features and employs a gated GNN and node pooling for improved classification and interpretability, achieved an accuracy of 87.2%. Alvi et al. [20] considered a Long Short-Term Memory (LSTM) for early detection of MCI using EEG signals. The model includes denoising, segmentation, down-sampling, and LSTM-based classification achieved an accuracy of 96.41% on a public EEG dataset. Amer and Belhaouari [9] developed a forward–backward fourier transform (FBFT) to convert EEG signals into time–frequency images. Various CNN models (GoogleNet, AlexNet, SqueezeNet) were used for classifying brain disorders including epilepsy, AD, murmur, and stress, achieving an accuracies of 99.82%, 95.91%, 85.1%, and 100%, respectively. Nour et al. [21] introduced a deep ensemble learning (DEL) model integrating five diverse 2D-CNN architectures without manual feature extraction. It achieved an average classification accuracy of 97.9%. LEADNet, a low-complexity CNN model, is introduced by Puri et al. [22] for the early detection of AD using spatiotemporal EEG signals,

TABLE I
EEG-BASED AD DETECTION METHODS PUBLISHED IN THE LITERATURE

Study	Model Type	Classes	Strengths	Limitations
Cassani and Falk, 2020 [10]	Modulation spectral patch features, SVM	AD vs. CN	Time-frequency modulation features; effective in detecting AD and its severity	Multi-stage pipeline limits real-time applicability
Li et al., 2021 [11]	VAE, Takagi-Sugeno-Kang fuzzy classifier	AD vs. CN	Captures latent dynamics across EEG and obtained better performance using energy in multiple frequency bands	Requires precise tuning of VAE and complex to interpret latent factor representation
Safi and Safi, 2021 [12]	EEG time-domain features (Hjorth parameters), ML classifiers (KNN, SVM, etc.)	AD vs. CN	Interpretable time-domain features; computationally efficient	Limited scalability; lacks deep modeling of EEG patterns; developed for binary classification
Cejnek et al., 2021 [13]	Channel-wise statistical measures, One-Class SVM	CN vs. MCI vs. AD	Effective for early diagnosis using reference class modeling	Depends on choice of reference class; hand-crafted features may miss deeper EEG patterns; lack of generalizability
Rodrigues et al., 2021 [14]	Lacunarity features, SVM, and ANN	AD vs. CN vs. MCI	Introduces lacsogram feature and multiband cepstral using DWT, ANN	Relies on hand-crafted features; less performance on ternary classification; lack of generalizability
Klepl et al., 2022 [15]	Functional connectivity matrices, GNN	AD vs. CN	Explores multiple brain connectivity methods; uses graph topology for improved interpretability	High computational cost; limited interpretability of learned GNN weights
Araújo et al., 2022 [16]	Classic ML (SVM, KNN, DT, etc.) and CNN	AD vs. CN vs. MCI	Rich nonlinear + statistical feature set using DWPT; interpretable regional electrode analysis	lower accuracy in multi-class setting (56.8%); CNN underperformed due to data size
Dogan et al., 2023 [17]	Graph-based handcrafted feature model with KNN classifier	AD vs. CN	Biologically-inspired feature extractor; multi-level feature generation; lightweight linear complexity	Evaluation restricted to binary class only; KNN may fail to perform better on high-dimensional data
Lopes et al., 2023 [18]	Modulation Spectrograms, CNN with Saliency Maps + SVM	AD vs. CN	Data-driven patch extraction; high interpretability	Lack of generalizability; limited performance on other datasets
Klepl et al., 2023 [19]	PSD features, AGGCN	AD vs. CN	Learns interpretable brain graphs; robust across eyes-open and eyes-closed	Small dataset size; relies only on PSD features; high model complexity with many hyper-parameters
Alvi et al., 2023 [20]	LSTM	AD vs. CN vs. MCI	Efficient pre-processing; time-efficient on CPU-only systems	Limited generalizability without external validation
Amer and Belhaouari, 2024 [9]	FBFT-based Images, CNN	AD vs. CN	FBFT transform improves interpretability; includes naked-eye visual analysis	Depends on pre-trained models; manual interpretation subject to human variability; high computational demand of FBFT and CNN
Nour et al., 2024 [21]	DEL with 2D-CNNs	AD vs. CN	No manual feature extraction; cross-validated robustness	Limited interpretability; tested on limited public datasets only
Puri et al., 2024 [22]	CNN (LEADNet)	AD vs. CN	Low complexity; superior to pre-trained models	Tested on a small dataset; limited subject diversity
Singh and Pachori, 2024 [23]	Improved MCh-EVDHM + ML classifiers	AD vs. CN	Rhythm separation using adaptive method; small model size (23–49 kB)	Requires multichannel EEG data; not tested for real-time deployment or larger populations
Sreedhar et al., 2025 [24]	Fuzzy RNN	AD vs. CN vs. FTD	Captures temporal dynamics; interpretable fuzzy rules	Requires high computation; limited generalizability without larger datasets
Lee and Lee, 2025 [25]	Classical ML (LDA, SVM, CatBoost, etc.) using Hjorth Parameters	AD vs. CN vs. FTD	Interpretable time-domain features; strong MMSE correlation ($r = 0.70$)	Limited dataset size; requires further validation for clinical generalizability; sensitive to EEG artifacts
Kalambe et al., 2025 [26]	2D Image from 8-sec segments, CNN (SBPFAN)	AD vs. CN vs. FTD vs. MCI	Satisfied performance across subject-dependent, independent, and cross-dataset setups; interpretable via t-SNE	Higher number of parameters; high computation cost

obtained a 99.24% accuracy. Singh and Pachori [23] presented a model consists of multichannel eigenvalue decomposition of hankel matrix (MCh-EVDHM) for rhythm separation. Features like energy, standard deviation, and skewness are extracted from EEG rhythms and classified using quadratic SVM. The model achieved 98.9% accuracy for eyes closed, 95.6% for eyes open, and 97.3% for combined states. Sreedhar et al. [24] developed a fuzzy recurrent neural network (RNN) model that uses EEG signals to categorize AD, FTD, and CN patients using sequential learning from RNNs and fuzzy logic. The model effectively captures temporal patterns and manages uncertainty in medical data. It achieved an accuracy up to 98.82% (AD vs CN), 98.42% (FTD vs CN), and 97.82% (AD vs FTD). Lee and Lee [25] introduced a EEG-based multiclass classification using time-domain Hjorth parameters. By applying optimal temporal segmentation (80s window), the model achieved classification accuracy: 92.5% for AD vs FTD vs CN, 88.6% for AD vs CN using LDA, and 96.0% for FTD vs CN using SVM. Kalambe et al. [26] introduces a separable bi-pyramidal feature attention network (SBPFAN) that extracts multiscale features using separable and dilated convolutions, enhanced by feature attention blocks that highlight disease-relevant patterns. SBPFAN achieved accuracy up

to 99.65% for AD vs. CN and 98.47% for AD vs. MCI vs. CN. Acharya et al. [27], presented a two-dimensional EEG-ConvNext to identify AD and achieved an accuracy of 97%. Sharma et al. [28], investigated the graph wavelet transform and graph Fourier transform features for the identification of AD. Dharia et al. [29], introduced a dual transformer cross attention network for classification of AD using 6 channel EEG. However, most of the existing methods suffer from high computational complexity, limited generalizability, reliance on hand-crafted features, and poor interpretability. In addition, most studies evaluated their methods using a single dataset, with some using larger private datasets. To address these challenges, the proposed approach incorporates three publicly available datasets to enhance generalizability.

III. METHODS & MATERIALS

This section outlines the methodology adopted in the development of LDANet for AD detection. The overall framework of our approach is illustrated in Fig. 1.

A. Dataset Description

Data collection is the first step in the proposed pipeline (see Fig. 1). In this phase, we collect three publicly available

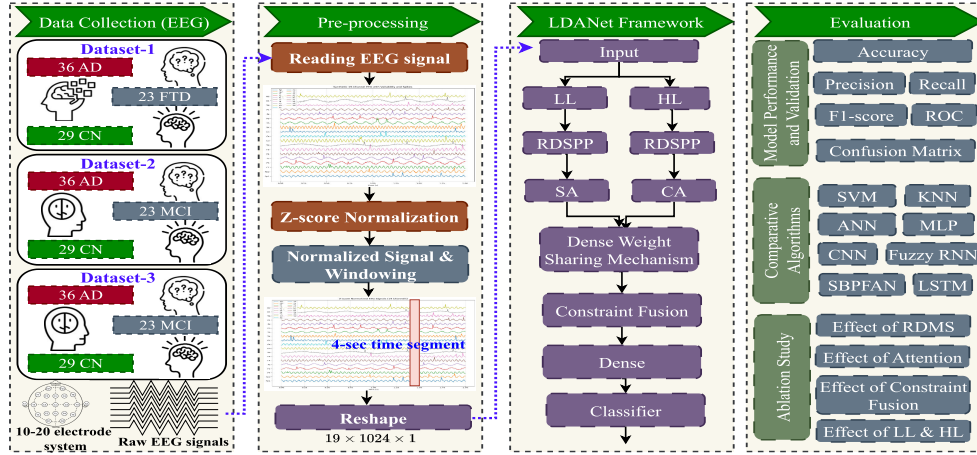


Fig. 1. General pipeline of the proposed approach.

TABLE II

THE DETAILED DESCRIPTION OF THREE PUBLICLY AVAILABLE EEG-BASED AD DATASETS

Dataset	Source	Channel & Sensor System	Class	Mean Age ($Years \pm sd$)	Count (Male/Female)
Dataset-1	openneuro.org	19, Nihon Kohden EEG 2100	AD	66.4 $\pm$ 7.9	36 (13/23)
			CN	67.9 $\pm$ 5.4	29 (11/18)
			FTD	63.6 $\pm$ 8.2	23 (14/9)
Dataset-2	figshare.com	19, Walter EEG PL-231	AD	70.5 $\pm$ 4.9	59 (28/31)
			CN	72.2 $\pm$ 5.3	102 (43/59)
			MCI	67 $\pm$ 7.6	7 (3/4)
Dataset-3	mendeley.org	4, Mitsar amplifier EEG	AD	75.31 $\pm$ 9.90	13 (5/8)
			CN	69.27 $\pm$ 6.65	15 (7/8)
			MCI	66.57 $\pm$ 6.85	7 (3/4)

benchmark datasets to assess the proposed LDANet. The international 10–20 electrode placement strategy was utilized across all datasets to standardize data collection. Dataset-1 was collected by the Department of Neurology at AHEPA General Hospital using a Nihon Kohden EEG 2100 device with 19 scalp electrodes at a sampling frequency of 500 Hz [30]. This dataset includes a study of 88 individuals diagnosed with AD, FTD, and CN. In Dataset-2, the Ethics Authority of the Faculty Hospital in Hradec Králové approved the use of 21 scalp electrodes with the Walter EEG PL-231 system at 256 Hz [13]. This dataset contains 168 subjects classified as AD, MCI, or CN based on the NINCDS-ADRDA criteria. Three of the 21 reference electrodes were excluded. Dataset-3 [31] consists of 44 participants diagnosed with CN, MCI, or AD. However, nine individuals were excluded from further analysis due to EEG acquisition problems, stroke, traumatic brain injury, olfactory dysfunction, or neurodegenerative conditions other than AD, such as Parkinson’s disease or multiple system atrophy. Among the remaining 35 subjects, 15 were CN, 7 had MCI, and 13 were diagnosed with AD. The dataset format is $4 \times 600 \times Num_trial$. Starting from the first column, the EEG channels are Fp1, Fz, Cz, and Pz. The second dimension contains 600 EEG samples captured at 200 Hz, spanning from one second before stimulus onset to two seconds afterward. The third dimension corresponds to the number of trials, which may vary by participant due to the removal of some trials during pre-processing. Table II provides an overview of the datasets. This study does not cover Datasets-1, -2, and -3 in detail, for further information please refer to the original articles.

B. Data Pre-Processing

Figure 1 presents pre-processing as the second phase of the proposed method. Initially, EEG recordings from Dataset-1 and Dataset-2 were resampled to a uniform frequency of 256 Hz using the MNE resampling function (`mne.io.Raw.resample()`) to ensure temporal consistency across datasets. Both datasets originally contained 19 electrodes following the international 10–20 system; only the common 19 channel positions are retained and reordered to maintain identical spatial alignment. Mainly, the pre-processing stage consists of two main steps: normalization of raw EEG signals and segmentation of the normalized signals. It is important to note that artifacts such as eye blinks, muscle activity, or electrical interference have already been removed by the original authors of the benchmark datasets, ensuring cleaner signals for downstream processing. To standardize the signal across subjects and channels, we apply z-score normalization, resulting in a transformed signal $\tilde{E}^{(s)}$ defined as $\tilde{E}^{(s)} = (E^{(s)} - \mu_s) / \sigma_s$, where μ_s and σ_s represent the mean and standard deviation of $E^{(s)}$, respectively. This step brings the signals into a canonical form with zero mean and unit variance, ensuring consistency across the dataset. Following normalization, the signal $\tilde{E}^{(s)}$ is segmented using a sliding window approach. Let Δt_w denote the window duration (set to 4-sec), and Δt_o the overlap (set to 3-sec). Given a sampling rate f_s , the number of samples per window is $L_w = \Delta t_w \cdot f_s$, and the stride between consecutive windows is $L_s = (\Delta t_w - \Delta t_o) \cdot f_s$. This segmentation yields a sequence of windowed segments $\{S_1, S_2, \dots, S_n\}$, where each segment $S_i \in \mathbb{R}^{e \times L_w}$ contains e EEG channels and L_w time samples. This windowing strategy offers several benefits: (1) it increases the quantity of training samples to combat overfitting, (2) it captures temporal dependencies within the EEG signal, and (3) overlapping helps reduce discontinuity effects and improves robustness without requiring explicit data augmentation [32].

C. Architecture of LDANet

The LDANet model is a lightweight dual-branch attention-based neural network designed for the classification of EEG

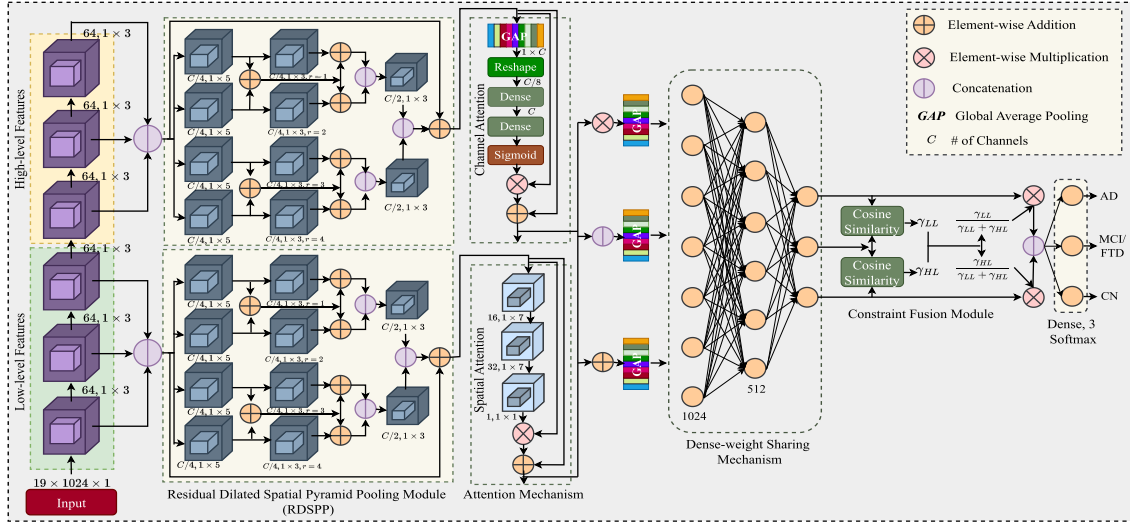


Fig. 2. The proposed LDANet for the identification of AD.

signals into three cognitive states: CN, MCI(or FTD), and AD. As illustrated in Fig. 2, the model processes input EEG segments with dimensions $19 \times 1024 \times 1$, where 19 corresponds to the number of EEG channels and 1024 denotes the number of time-domain samples within each segment window. While conventional methods primarily rely on CNNs to extract HL spatial-temporal features, they often focus on these representations in isolation, neglecting the complementary LL information that can enhance model interpretability and robustness. In contrast, LDANet introduces a more comprehensive architecture that captures feature representations at multiple abstraction levels. The model architecture is composed of three core modules: (1) a multilevel joint feature extraction framework, (2) a dual-branch attention mechanism, and (3) a constraint-based feature fusion strategy. The multilevel joint learning component simultaneously extracts both LL and HL features and incorporates a residual-dilated spatial pyramid pooling (RDSPP) module. This dual-branch setup enables the model to learn and preserve detailed temporal patterns from early layers while abstracting semantic features from deeper layers. Let us denote the input EEG segment as $X \in \mathbb{R}^{C \times T \times 1}$, where C is the number of channels and T is the number of time samples. The early-stage convolutional layers (e.g., first three layers of VGG16 or ResNet) extract LL features, represented as $\Upsilon_{LL} = \phi_{low}(X)$, where $\phi_{low}(\cdot)$ refers to the set of LL convolutional transformations. These features capture fine-grained patterns such as edges, bursts, or localized signal fluctuations. On the other hand, deeper layers in the network extract HL abstractions, modeled as $\Upsilon_{HL} = \phi_{high}(X')$, where $\phi_{high}(\cdot)$ denotes the transformation applied through the final layers of the backbone network. These features are more semantically meaningful and can represent class-specific attributes useful for diagnosis. LDANet's LL and HL feature extraction paths are mathematically defined in Eq. 1 & 2. The LL pathway captures fine-grained temporal patterns by passing the input EEG signal $X \in \mathbb{R}^{C \times T \times 1}$ through three sequential convolutional layers with kernel size 1×3 : $\ell_1 = \text{Conv}_{1 \times 3}^{(1)}(X)$, $\ell_2 = \text{Conv}_{1 \times 3}^{(2)}(\ell_1)$, $\ell_3 = \text{Conv}_{1 \times 3}^{(3)}(\ell_2)$.

The final LL feature map is then defined by channel-wise concatenating ($\parallel$) the intermediate outputs as shown in Eq. 1.

$$\Upsilon_{LL} = (\ell_1 \parallel \ell_2 \parallel \ell_3) \quad (1)$$

HL features are extracted by feeding the output ℓ_3 into a deeper convolutional stack: $h_1 = \text{Conv}_{1 \times 3}^{(4)}(\ell_3)$, $h_2 = \text{Conv}_{1 \times 3}^{(5)}(h_1)$, $h_3 = \text{Conv}_{1 \times 3}^{(6)}(h_2)$. The concatenated result forms the HL feature representation shown in Eq. 2.

$$\Upsilon_{HL} = (h_1 \parallel h_2 \parallel h_3) \quad (2)$$

The extracted LL & HL features are further processed by the RDSPP module to enhance the contextual depth and multi-scale representations. To fuse these dual-branch features, LDANet employs a constraint fusion module (CFM). The fused features are then passed through a softmax layer, producing classification probabilities across three cognitive states: CN, MCI (or FTD), and AD. Each convolutional layer in the LDANet architecture is followed by batch normalization to improve training stability and convergence speed; and a ReLU activation function is applied to introduce non-linearity, as illustrated in Fig. 2.

1) *Residual Dilated Spatial Pyramid Pooling*: Deep learning based models have shown considerable success across domains such as computer vision [33] and brain-computer interface [26], [32]. Since, traditional feature extractors capture information from a fixed scale, leading to a lack of sensitivity to multi-granular details. To address this, LDANet employs a RDSPP module to capture multi-scale contextual features. Four parallel descriptors $\Phi_i (i = 1, 2, 3, 4)$ are generated from both Υ_{LL} and Υ_{HL} through distinct 1×5 convolutions, each reducing the channel dimensionality to $\frac{1}{4}$ of the input. These are then processed through dilated convolutions as shown in Eq. 3.

$$\Delta_i = \mathcal{D}_{1 \times 3}^{(r_i)}(\Phi_i), \quad r_i = 2^{i-1}, \quad i \in \{1, 2, 3, 4\} \quad (3)$$

where $\mathcal{D}_{1 \times 3}^{(r)}(\cdot)$ denotes a dilated convolution with rate r and kernel size 1×3 , and Δ_i represents the output of the i^{th} dilation

path. To enable local-global residual enhancement, we perform element-wise addition ($\oplus$) between the dilated features and their paired source branches as represented in Eq. 4.

$$\begin{cases} \tilde{\Delta}_i = \Delta_i \oplus \Phi_1 \oplus \Phi_2, & i = 1, 2 \\ \tilde{\Delta}_i = \Delta_i \oplus \Phi_3 \oplus \Phi_4, & i = 3, 4 \end{cases} \quad (4)$$

To compress adjacent residual branches, we concatenate and project them via 1×1 convolutional filters depicted in Eq. 5.

$$\begin{cases} \Gamma_1 = \text{Conv}_{C/2, 1 \times 3}(\tilde{\Delta}_1 \parallel \tilde{\Delta}_2) \\ \Gamma_2 = \text{Conv}_{C/2, 1 \times 3}(\tilde{\Delta}_3 \parallel \tilde{\Delta}_4) \end{cases} \quad (5)$$

The combined features are aggregated to produce the fused multi-scale output and residual connectivity is then applied to preserve gradient flow and representation fidelity as shown in Eq. 6.

$$\Psi_{\text{out}} = \Upsilon_{\text{LL}} \oplus ((\Gamma_1 \parallel \Gamma_2)) \quad (6)$$

This RDSPP module is applied to the characteristics of LL (Υ_{LL}) and HL (Υ_{HL}), and obtained the output feature map denoted as $\Psi_{\text{out}}^{\text{LL}}$ & $\Psi_{\text{out}}^{\text{HL}}$.

2) *Attention Mechanism*: Directly combining multi-scale representations of RDSPP module without discrimination may introduce redundancy. More critically, irrelevant or noisy features at certain levels can degrade performance or lead to misclassification. Therefore, it is essential to selectively emphasize the most informative features [34]. Therefore, spatial attention (SA) and channel attention (CA) mechanisms are employed in LDANet, as detailed below.

a) *Spatial Attention*: The SA module in LDANet enhances the model's ability to focus on *where* discriminative regions appear within intermediate feature maps. Given the RDSPP output $\Psi_{\text{out}}^{\text{LL}} \in \mathbb{R}^{C \times T \times F}$, the SA module processes it through three sequential convolutional layers with kernel sizes 1×7 , 1×7 , and 1×1 , followed by a sigmoid activation function (σ) to produce a normalized SA mask $\mathcal{A}_{\text{spatial}} \in [0, 1]^{1 \times H \times W}$. The mask is computed as $\mathcal{A}_{\text{spatial}} = \sigma(\text{Conv}_{1 \times 1}^{(3)}(\text{Conv}_{1 \times 7}^{(2)}(\text{Conv}_{1 \times 7}^{(1)}(\Psi_{\text{out}}^{\text{LL}}))))$. This attention map is then broadcasted and element-wise multiplied ($\otimes$) with the original feature map to emphasize informative spatial locations, resulting in the refined feature $\Psi_{\text{SA}}^{\text{LL}} = \Psi_{\text{out}}^{\text{LL}} \otimes \mathcal{A}_{\text{spatial}}$. To preserve the original spatial structure and maintain gradient flow, a residual connection is applied as represented in Eq. 7

$$\tilde{\Psi}_{\text{SA}}^{\text{LL}} = \Psi_{\text{out}}^{\text{LL}} \oplus \Psi_{\text{SA}}^{\text{LL}} \quad (7)$$

b) *Channel-Wise Attention*: While CNNs encode diverse semantic cues across channels, not all contribute equally to classification. The CA mechanism in LDANet highlights channels strongly correlated with Alzheimer's-specific patterns by applying a CA module to the HL feature map $\Psi_{\text{out}}^{\text{HL}} \in \mathbb{R}^{C \times T \times F}$, obtained from the RDSPP block. To compute inter-channel dependencies, GAP is first applied to $\Psi_{\text{out}}^{\text{HL}}$ to obtain a descriptor $z \in \mathbb{R}^C$, where each element represents the average activation per channel. This vector is passed through two fully connected layers: the first reduces the dimensionality to $C/8$, followed by a ReLU activation; the second restores it to C , followed by a sigmoid function to generate the final attention vector $\mathcal{A}_{\text{channel}} \in [0, 1]^C$. The attention vector is then

broadcasted and applied to the original features via element-wise multiplication: $\tilde{\Psi}_{\text{CA}}^{\text{HL}} = \Psi_{\text{out}}^{\text{HL}} \otimes \mathcal{A}_{\text{channel}}$. To retain the original channel semantics, a residual connection is added as shown in Eq. 8.

$$\tilde{\Psi}_{\text{CA}}^{\text{HL}} = \Psi_{\text{out}}^{\text{HL}} \oplus \Psi_{\text{CA}}^{\text{HL}} \quad (8)$$

After the attention mechanism, three operations ($\oplus, \otimes, \parallel$) are applied to integrate the enhanced LL ($\tilde{\Psi}_{\text{SA}}^{\text{LL}}$) and HL ($\tilde{\Psi}_{\text{CA}}^{\text{HL}}$) features. To reduce the network's computational overhead while retaining semantic richness, GAP is applied to each resulting tensor. Specifically, we define $\Omega_{\oplus} = \text{GAP}(\tilde{\Psi}_{\text{SA}}^{\text{LL}} \oplus \tilde{\Psi}_{\text{CA}}^{\text{HL}})$, $\Omega_{\otimes} = \text{GAP}(\tilde{\Psi}_{\text{SA}}^{\text{LL}} \otimes \tilde{\Psi}_{\text{CA}}^{\text{HL}})$, and $\Omega_{\parallel} = \text{GAP}(\tilde{\Psi}_{\text{SA}}^{\text{LL}} \parallel \tilde{\Psi}_{\text{CA}}^{\text{HL}})$.

3) *Dense-Weight Sharing Mechanism*: The dense weight-sharing mechanism (DWSM) in LDANet utilizes a two-layer fully connected structure shared across three distinct feature vectors— $\Omega_{\oplus}$, $\Omega_{\otimes}$, and $\Omega_{\parallel}$. Each input first passes through an independent 512-unit dense layer with ReLU activation. The resulting vectors are then processed by two sequential dense layers with shared weights and biases, denoted W_1 , b_1 and W_2 , b_2 , respectively. The transformation is applied as $z_i = \varsigma_2(W_2 \cdot \varsigma_1(W_1 \cdot \Omega_i + b_1) + b_2)$, where ς_1 and ς_2 represent ReLU activation functions, and $i \in \{\oplus, \otimes, \parallel\}$. This weight-sharing design promotes representational consistency and reduces parameter overhead across branches. The mathematical representation is displayed in Eq. 9.

$$\begin{cases} \mathcal{A}_{\oplus} = \text{DWSM}(\Omega_{\oplus}) = \varsigma(W_s \cdot \Omega_{\oplus} + b_s) \\ \mathcal{M}_{\otimes} = \text{DWSM}(\Omega_{\otimes}) = \varsigma(W_s \cdot \Omega_{\otimes} + b_s) \\ \mathcal{C}_{\parallel} = \text{DWSM}(\Omega_{\parallel}) = \varsigma(W_s \cdot \Omega_{\parallel} + b_s) \end{cases} \quad (9)$$

where $\text{DWSM}(\bullet)$ denotes the dense layer weight-sharing mechanism, W_s and b_s indicates the trainable weights and bias of the weight-sharing dense layers, respectively.

4) *Constraint Fusion Module*: LL features capture localized discriminative patterns, while HL features encode global semantics. Although their fusion enhances recognition robustness, assigning equal importance to both may degrade performance, especially when one branch contributes less under specific sample conditions. To address this, we integrate a CFM [35], enabling adaptive weighting of $\mathcal{A}_{\oplus}$ and $\mathcal{M}_{\otimes}$ based on their relevance. The similarity of $\mathcal{A}_{\oplus}$ and $\mathcal{M}_{\otimes}$ to the direct fusion feature $\mathcal{C}_{\parallel}$ is measured via cosine similarity, defined in Eq. 10.

$$\begin{cases} \gamma_{\text{LL}} = \mathcal{C}(\mathcal{A}_{\oplus}, \mathcal{C}_{\parallel}) \\ \gamma_{\text{HL}} = \mathcal{C}(\mathcal{M}_{\otimes}, \mathcal{C}_{\parallel}) \end{cases} \quad (10)$$

where $\mathcal{C}(P, Q) = \frac{\sum_{j=1}^n p_j q_j}{\sqrt{\sum_{j=1}^n p_j^2} \sqrt{\sum_{j=1}^n q_j^2}}$ denotes cosine similarity, and p_j , q_j are the j^{th} elements of vectors P and Q . Cosine scores are normalized and converted into attention weights: $\beta_{\text{LL}} = \frac{\gamma_{\text{LL}}}{\gamma_{\text{LL}} + \gamma_{\text{HL}}}$ and $\beta_{\text{HL}} = \frac{\gamma_{\text{HL}}}{\gamma_{\text{LL}} + \gamma_{\text{HL}}}$. These are applied to adaptively fuse features as represented in Eq. 11.

$$\text{CFM} = (\beta_{\text{LL}} \otimes \mathcal{A}_{\oplus}) \parallel (\beta_{\text{HL}} \otimes \mathcal{M}_{\otimes}) \quad (11)$$

This constraint-guided fusion dynamically balances LL and HL contributions, improving the model's ability to prioritize the most informative features under varying conditions. Finally, a softmax layer is applied to obtain classification probabilities.

IV. EXPERIMENTAL RESULTS

This section presents an overview of the experimental configurations, including details on parameter settings, execution, and measurement metrics.

A. Implementation Details

To implement the proposed LDANet in Keras framework, several hyperparameters are carefully configured. Categorical cross-entropy is used as the loss function. In the proposed framework, all convolutional and fully connected layers were initialized using the Glorot (Xavier) uniform initializer, while bias terms were initialized to zero. The batch normalization scale and shift parameters (γ , β) were initialized to one and zero, respectively. The network was optimized using the Adam optimizer with a fixed learning rate of 1×10^{-3} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-7}$. Training is conducted with a batch size of 16 for up to 200 epochs. An early stopping strategy (patience = 5, monitored on validation loss) is applied to prevent overfitting and automatically restore the best-performing model weights. To ensure reproducibility, all random seeds were fixed across the NumPy and TensorFlow environments. Three benchmark AD datasets are used for evaluation: Dataset-1 (CN vs. AD, CN vs. FTD, CN vs. FTD vs. AD), Dataset-2 (CN vs. AD, CN vs. MCI, CN vs. MCI vs. AD), and Dataset-3 (CN vs. AD, CN vs. MCI, CN vs. MCI vs. AD). Tenfold cross-validation is applied, each fold using 80% of the data for training and 20% for validation & testing to ensure robust and generalized results.

B. Evaluation Metrics

The effectiveness of LDANet is evaluated using standard performance metrics derived from the confusion matrix: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). From these values, several key metrics are computed. Accuracy is defined as $\mathcal{A}(\%) = \frac{TP+TN}{TP+TN+FP+FN} \times 100$, reflecting the overall prediction correctness. Precision and recall, represented as $\mathcal{P}(\%) = \frac{TP}{TP+FP}$ and $\mathcal{R}(\%) = \frac{TP}{TP+FN}$, respectively, quantify the model's ability to avoid false positives and false negatives. The F1-score, written as $\mathcal{F}(\%) = \frac{2 \cdot \mathcal{P} \cdot \mathcal{R}}{\mathcal{P} + \mathcal{R}} \times 100$, combines precision and recall in a single measure. Additionally, the Receiver Operating Characteristic (ROC) curve is used to assess class separation, with the Area Under the Curve (AUC) indicating this ability: 50% reflects random performance, and 100% indicates perfect distinction. These metrics provide a comprehensive evaluation of LDANet compared to thirteen SOTA approaches. The performance metrics are reported as mean $\pm$ standard deviation ($X \pm Y$) across tenfold evaluations, representing average accuracy and its dispersion across different splits.

C. Performance Analysis

The results of the proposed LDANet on Dataset-1, Dataset-2, and Dataset-3 are encapsulated in Fig. 3, which illustrates the mean normalized confusion matrices. In Fig. 3, the rows represent the true labels in the test data, while the columns indicate the predicted labels. The diagonal elements of the

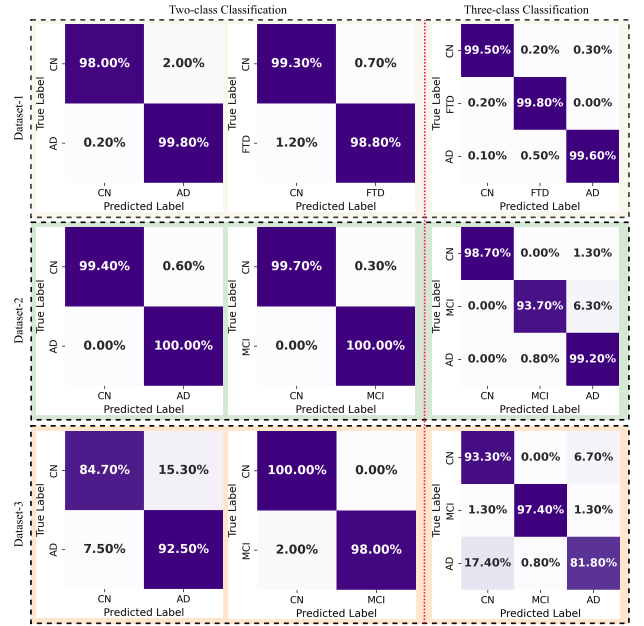


Fig. 3. Confusion matrices of the proposed LDANet on Dataset-1, -2, and -3.

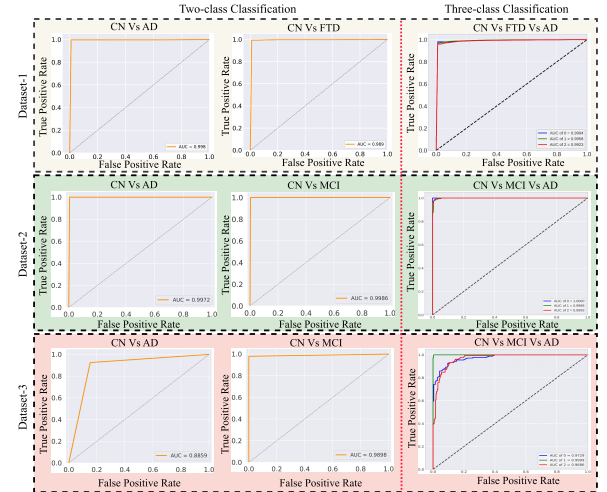


Fig. 4. ROC curves with AUC values for LDANet across datasets 1, 2, and 3 for subject-dependent evaluation.

matrix represent the normalized scores of the correctly identified categories. The AUC-ROC curve is a widely utilized statistic for assessing the effectiveness of the proposed method. The model's performance improves with an increase in AUC. Figure 4 illustrates the AUC-ROC curves for datasets-1, -2, and -3. We derive insights at the dataset level from Figures 3 and 4 as follows:

1) *Dataset-1*: Three distinct evaluations were conducted on Dataset-1: CN vs. AD, CN vs. FTD, and CN vs. FTD Vs AD. The average class-wise confusion matrix of the proposed LDANet on Dataset-1 is shown in Fig. 3 (first row) and AUC-ROC curve is depicted in Fig. 4 (first row). Figure 3 demonstrates that the proposed LDANet obtained 98% accurate prediction rate for CN and 99.8% for AD in two-class classification of CN vs. AD. Similarly, for CN vs. FTD the

TABLE III
COMPARISON OF THE LDANet WITH SOTA APPROACHES USING DATASET-1 BASED ON CLASSIFICATION PERFORMANCE

Method	Dataset-1											
	CN vs. AD				CN vs. FTD				CN vs. FTD vs. AD			
	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$
LCADNet [36]	0.9392 ± 0.011	0.93 ± 0.018	0.93 ± 0.031	0.93 ± 0.014	0.9486 ± 0.024	0.96 ± 0.033	0.94 ± 0.010	0.95 ± 0.016	0.9339 ± 0.075	0.95 ± 0.068	0.93 ± 0.057	0.94 ± 0.018
AD-Hjorth [12]	0.945 ± 0.018	0.950 ± 0.021	0.940 ± 0.020	0.945 ± 0.019	0.952 ± 0.016	0.957 ± 0.018	0.948 ± 0.017	0.952 ± 0.017	0.931 ± 0.022	0.936 ± 0.020	0.928 ± 0.021	0.932 ± 0.021
VAE-FZ [11]	0.8913 ± 0.016	0.89 ± 0.033	0.89 ± 0.016	0.89 ± 0.018	0.9106 ± 0.057	0.91 ± 0.061	0.91 ± 0.043	0.91 ± 0.033	0.8864 ± 0.071	0.89 ± 0.082	0.88 ± 0.018	0.88 ± 0.071
GNN [15]	0.9518 ± 0.011	0.95 ± 0.016	0.95 ± 0.012	0.95 ± 0.017	0.9738 ± 0.014	0.97 ± 0.018	0.97 ± 0.011	0.97 ± 0.014	0.9482 ± 0.021	0.95 ± 0.023	0.97 ± 0.011	0.95 ± 0.032
ML-CNN [16]	0.9016 ± 0.035	0.91 ± 0.054	0.89 ± 0.076	0.90 ± 0.054	0.9231 ± 0.031	0.92 ± 0.046	0.92 ± 0.038	0.92 ± 0.066	0.8993 ± 0.036	0.90 ± 0.031	0.90 ± 0.032	0.90 ± 0.033
AGGCN [19]	0.7781 ± 0.154	0.78 ± 0.126	0.80 ± 0.081	0.75 ± 0.102	0.7365 ± 0.163	0.63 ± 0.16	0.78 ± 0.127	0.72 ± 0.162	0.7562 ± 0.147	0.75 ± 0.133	0.75 ± 0.126	0.75 ± 0.164
DEL-AD [21]	0.8148 ± 0.128	0.82 ± 0.103	0.81 ± 0.111	0.80 ± 0.118	0.7312 ± 0.116	0.73 ± 0.111	0.73 ± 0.124	0.73 ± 0.167	0.8019 ± 0.166	0.80 ± 0.176	0.80 ± 0.133	0.80 ± 0.101
EEGConvNeXt [27]	0.9136 ± 0.011	0.91 ± 0.018	0.91 ± 0.035	0.91 ± 0.011	0.9461 ± 0.018	0.95 ± 0.017	0.95 ± 0.021	0.95 ± 0.026	0.9266 ± 0.046	0.93 ± 0.053	0.93 ± 0.056	0.93 ± 0.048
MMSDL [37]	0.9002 ± 0.026	0.87 ± 0.031	0.89 ± 0.034	0.88 ± 0.011	0.8971 ± 0.013	0.89 ± 0.010	0.89 ± 0.033	0.89 ± 0.021	0.8788 ± 0.038	0.87 ± 0.024	0.93 ± 0.011	0.89 ± 0.012
edRVFL [38]	0.9691 ± 0.013	0.97 ± 0.015	0.97 ± 0.005	0.97 ± 0.009	0.9491 ± 0.035	0.95 ± 0.027	0.95 ± 0.021	0.95 ± 0.024	0.9762 ± 0.011	0.97 ± 0.015	0.97 ± 0.013	0.97 ± 0.011
DCNN-SBIL [39]	0.9591 ± 0.011	0.96 ± 0.032	0.96 ± 0.026	0.96 ± 0.018	0.9396 ± 0.062	0.93 ± 0.019	0.95 ± 0.028	0.94 ± 0.011	0.9219 ± 0.0412	0.92 ± 0.061	0.92 ± 0.038	0.92 ± 0.051
SBPFA [26]	0.9809 ± 0.003	0.99 ± 0.004	0.98 ± 0.009	0.99 ± 0.004	0.9868 ± 0.004	0.99 ± 0.006	0.97 ± 0.015	0.99 ± 0.004	0.9954 ± 0.002	0.99 ± 0.003	0.99 ± 0.007	0.99 ± 0.008
Hjorth-SVM [25]	0.886 ± 0.012	0.882 ± 0.018	0.921 ± 0.017	0.900 ± 0.016	0.894 ± 0.015	0.889 ± 0.019	0.918 ± 0.018	0.903 ± 0.017	0.876 ± 0.021	0.872 ± 0.020	0.885 ± 0.019	0.879 ± 0.018
LDANet	0.989 ± 0.002	0.99 ± 0.001	0.99 ± 0.0013	0.99 ± 0.0012	0.9905 ± 0.004	0.99 ± 0.0033	0.99 ± 0.001	0.99 ± 0.001	0.9963 ± 0.0011	0.99 ± 0.0012	0.99 ± 0.001	0.99 ± 0.001

TABLE IV
COMPARISON OF THE LDANet WITH SOTA APPROACHES USING DATASET-2 BASED ON CLASSIFICATION PERFORMANCE

Method	Dataset-2											
	CN vs. AD				CN vs. MCI				CN vs. MCI vs. AD			
	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$
LCADNet [36]	0.9029 ± 0.025	0.89 ± 0.101	0.90 ± 0.032	0.89 ± 0.016	0.9162 ± 0.023	0.85 ± 0.051	0.92 ± 0.021	0.91 ± 0.041	0.8815 ± 0.027	0.88 ± 0.028	0.85 ± 0.047	0.86 ± 0.038
AD-Hjorth [12]	0.953 ± 0.017	0.958 ± 0.020	0.946 ± 0.019	0.952 ± 0.018	0.961 ± 0.015	0.965 ± 0.017	0.951 ± 0.016	0.957 ± 0.016	0.938 ± 0.021	0.942 ± 0.020	0.934 ± 0.019	0.938 ± 0.019
VAE-FZ [11]	0.860 ± 0.025	0.89 ± 0.013	0.86 ± 0.018	0.88 ± 0.011	0.9033 ± 0.021	0.91 ± 0.032	0.90 ± 0.033	0.90 ± 0.026	0.8765 ± 0.051	0.88 ± 0.022	0.88 ± 0.028	0.88 ± 0.021
GNN [15]	0.9184 ± 0.031	0.92 ± 0.027	0.92 ± 0.029	0.92 ± 0.031	0.9516 ± 0.018	0.97 ± 0.011	0.95 ± 0.024	0.96 ± 0.011	0.9277 ± 0.013	0.93 ± 0.016	0.93 ± 0.011	0.93 ± 0.028
ML-CNN [16]	0.9233 ± 0.018	0.93 ± 0.011	0.91 ± 0.033	0.92 ± 0.013	0.9411 ± 0.019	0.94 ± 0.011	0.94 ± 0.023	0.94 ± 0.016	0.9368 ± 0.036	0.94 ± 0.011	0.94 ± 0.021	0.94 ± 0.014
AGGCN [19]	0.8818 ± 0.092	0.85 ± 0.084	0.85 ± 0.073	0.85 ± 0.081	0.8279 ± 0.039	0.83 ± 0.041	0.83 ± 0.073	0.83 ± 0.61	0.8009 ± 0.084	0.80 ± 0.073	0.82 ± 0.061	0.81 ± 0.083
DEL-AD [21]	0.8148 ± 0.128	0.82 ± 0.103	0.81 ± 0.111	0.80 ± 0.118	0.7312 ± 0.116	0.73 ± 0.111	0.73 ± 0.124	0.73 ± 0.167	0.8019 ± 0.166	0.80 ± 0.176	0.80 ± 0.133	0.80 ± 0.101
EEGConvNeXt [27]	0.9518 ± 0.011	0.96 ± 0.032	0.96 ± 0.026	0.96 ± 0.018	0.9396 ± 0.062	0.93 ± 0.019	0.95 ± 0.028	0.94 ± 0.011	0.9219 ± 0.0412	0.92 ± 0.061	0.92 ± 0.038	0.92 ± 0.051
MMSDL [37]	0.9612 ± 0.025	0.96 ± 0.015	0.96 ± 0.016	0.96 ± 0.018	0.9324 ± 0.011	0.95 ± 0.009	0.93 ± 0.008	0.93 ± 0.011	0.9566 ± 0.024	0.896 ± 0.019	0.96 ± 0.013	0.96 ± 0.015
edRVFL [38]	0.9676 ± 0.012	0.97 ± 0.011	0.97 ± 0.016	0.97 ± 0.001	0.9613 ± 0.009	0.96 ± 0.008	0.96 ± 0.011	0.96 ± 0.016	0.9710 ± 0.018	0.97 ± 0.011	0.97 ± 0.013	0.97 ± 0.012
DCNN-SBIL [39]	0.9487 ± 0.028	0.95 ± 0.031	0.95 ± 0.026	0.95 ± 0.022	0.9613 ± 0.024	0.95 ± 0.022	0.97 ± 0.028	0.96 ± 0.031	0.9552 ± 0.018	0.96 ± 0.011	0.96 ± 0.013	0.96 ± 0.026
SBPFA [26]	0.9965 ± 0.001	0.99 ± 0.008	0.99 ± 0.005	0.99 ± 0.005	0.9908 ± 0.002	0.99 ± 0.002	0.98 ± 0.001	0.99 ± 0.006	0.9847 ± 0.003	0.96 ± 0.003	0.99 ± 0.001	0.97 ± 0.002
Hjorth-SVM [25]	0.892 ± 0.014	0.888 ± 0.019	0.918 ± 0.018	0.903 ± 0.017	0.901 ± 0.012	0.896 ± 0.018	0.924 ± 0.017	0.909 ± 0.016	0.883 ± 0.020	0.879 ± 0.021	0.894 ± 0.019	0.886 ± 0.018
LDANet	0.9998 ± 0.0001	1.0 ± 0.00	1.0 ± 0.00	1.0 ± 0.00	0.9982 ± 0.0011	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.9835 ± 0.0023	0.98 ± 0.0062	0.97 ± 0.0018	0.98 ± 0.0013

proposed approach obtained 99.3% for CN and 98.8% in FTD. For three-class classification, the proposed LDANet achieved an accurate detection rate of CN is 99.5%, FTD is 99.8%, and AD is 99.6%. The FP rate of the proposed LDANet in two-class classification ranges from 0.2% to 1.2%. Similarly, for three-class classification the FP rate ranges from 0 to 0.5%. The efficacy of the proposed LDANet is demonstrated by the AUC score of 0.998 for CN vs. AD, 0.989 for CN vs. FTD, and for three-class classification, class-0, class-1, and class-2 achieved a score of 0.9994, 0.9958, and 0.9923, respectively. Here, class-0 indicates CN, class-1 represents FTD, class-2 denote AD. From the confusion matrix given in Fig. 3, $\mathcal{A}$, $\mathcal{P}$, $\mathcal{R}$, and $\mathcal{F}$ are measured to evaluate the proposed LDANet, results are reported in Table III.

2) *Dataset-2*: Three evaluation settings were applied to Dataset-2: CN vs. AD, CN vs. MCI, and CN vs. MCI vs. AD. The average class-wise confusion matrix and AUC-ROC curves for these tasks are presented in Fig. 3 and Fig. 4 (second row). For CN vs. AD, LDANet achieved 99.4% detection rate for CN and 100% for AD. For CN vs. MCI, 99.7% for CN and 100% for MCI. In the three-class setting, the model reached 98.7% for CN, 93.7% for MCI, and 99.2% for AD. The FP rate ranged from 0 to 0.6% in binary tasks and up to 6.3% in the three-class setting. AUC scores further highlight the model's performance: 0.9972 for CN vs. AD, 0.9986 for CN vs. MCI, and in the three-class setup, 1.0 for class-0 (CN), 0.9988 for class-1 (MCI), and 0.9993 for class-2 (AD), as shown in Fig. 4. Evaluation metrics including $\mathcal{A}$, $\mathcal{P}$, $\mathcal{R}$, and $\mathcal{F}$ are summarized in Table IV.

3) *Dataset-3*: Three evaluation scenarios were conducted on Dataset-3: CN vs. AD, CN vs. MCI, and CN vs. MCI vs. AD. The corresponding average confusion matrices and AUC-ROC curves are shown in Fig. 3 and Fig. 4 (third row).

In the CN vs. AD task, LDANet achieved detection rates of 84.07% for CN and 92.5% for AD. For CN vs. MCI, the model reached 100% for CN and 98% for MCI. Under the three-class setting, classification accuracies are 93.3% for CN, 97.4% for MCI, and 81.8% for AD. The FP rate varied from 0 to 15.0% in binary classification, and reached up to 17.4% in the three-class task. AUC scores further validate model performance: 0.8859 for CN vs. AD, 0.9898 for CN vs. MCI, and for the three-class case, 0.9729 for class-0 (CN), 0.9999 for class-1 (MCI), and 0.9686 for class-2 (AD), as depicted in Fig. 4. Evaluation metrics including $\mathcal{A}$, $\mathcal{P}$, $\mathcal{R}$, and $\mathcal{F}$ are detailed in Table V. Although LDANet performs competitively on Dataset-3, a lower AUC is observed for the CN vs AD task. This outcome is mainly due to the limited number of subjects and the use of only four EEG channels, which restricts AD-relevant information and reduces the model's ability to capture discriminative Alzheimer's-related patterns. With just 28 subjects (15 CN and 13 AD), the small cohort also increases fold-to-fold variability during cross-validation. Moreover, an exceptionally high accuracy in the CN vs MCI task (99.2%) compared to CN vs AD (88.6%) is observed due to the small and homogeneous MCI group ($n = 7$), whose consistent EEG characteristics make class separation appear easier. In contrast, the AD group shows greater variability across subjects, lowering its overall separability. Despite these constraints, LDANet consistently outperforms all baseline methods, demonstrating strong robustness even under data-limited conditions.

4) *Comparison With SOTA Approaches*: The proposed LDANet is evaluated against 13 SOTA methods under identical experimental conditions across three datasets. The results are presented in Table III for Dataset-1, Table IV for Dataset-2, and Table V for Dataset-3. These tables also report key performance metrics, including $\mathcal{A}$, $\mathcal{P}$, $\mathcal{R}$, and $\mathcal{F}$. Based on

TABLE V
COMPARISON OF THE LDANet WITH SOTA APPROACHES USING DATASET-3 BASED ON CLASSIFICATION PERFORMANCE

Method	Dataset-3											
	CN vs. AD				CN vs. MCI				CN vs. MCI vs. AD			
	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$
LCADNet [36]	0.7218 $\pm$ 0.103	0.740 $\pm$ 0.120	0.700 $\pm$ 0.142	0.710 $\pm$ 0.111	0.7892 $\pm$ 0.082	0.790 $\pm$ 0.094	0.800 $\pm$ 0.085	0.790 $\pm$ 0.090	0.7634 $\pm$ 0.096	0.760 $\pm$ 0.095	0.760 $\pm$ 0.093	0.760 $\pm$ 0.091
AD-Hjorth [12]	0.814 $\pm$ 0.024	0.818 $\pm$ 0.026	0.808 $\pm$ 0.025	0.813 $\pm$ 0.024	0.832 $\pm$ 0.022	0.838 $\pm$ 0.024	0.826 $\pm$ 0.023	0.832 $\pm$ 0.022	0.796 $\pm$ 0.027	0.801 $\pm$ 0.028	0.790 $\pm$ 0.026	0.795 $\pm$ 0.026
VAE-FZ [11]	0.6761 $\pm$ 0.124	0.670 $\pm$ 0.101	0.680 $\pm$ 0.135	0.670 $\pm$ 0.127	0.7032 $\pm$ 0.089	0.700 $\pm$ 0.079	0.700 $\pm$ 0.083	0.700 $\pm$ 0.091	0.7173 $\pm$ 0.108	0.720 $\pm$ 0.106	0.720 $\pm$ 0.097	0.720 $\pm$ 0.101
GNN [15]	0.7697 $\pm$ 0.084	0.780 $\pm$ 0.093	0.750 $\pm$ 0.101	0.760 $\pm$ 0.091	0.8419 $\pm$ 0.045	0.860 $\pm$ 0.062	0.830 $\pm$ 0.053	0.830 $\pm$ 0.051	0.7356 $\pm$ 0.095	0.740 $\pm$ 0.098	0.740 $\pm$ 0.095	0.740 $\pm$ 0.093
ML-CNN [16]	0.7342 $\pm$ 0.093	0.730 $\pm$ 0.086	0.740 $\pm$ 0.083	0.730 $\pm$ 0.090	0.8001 $\pm$ 0.074	0.800 $\pm$ 0.062	0.800 $\pm$ 0.068	0.800 $\pm$ 0.070	0.7516 $\pm$ 0.082	0.750 $\pm$ 0.079	0.750 $\pm$ 0.078	0.750 $\pm$ 0.080
AGGCN [19]	0.6864 $\pm$ 0.139	0.690 $\pm$ 0.131	0.680 $\pm$ 0.142	0.680 $\pm$ 0.130	0.7679 $\pm$ 0.092	0.770 $\pm$ 0.091	0.770 $\pm$ 0.086	0.770 $\pm$ 0.091	0.6910 $\pm$ 0.132	0.700 $\pm$ 0.120	0.690 $\pm$ 0.128	0.690 $\pm$ 0.127
DEL-AD [21]	0.6915 $\pm$ 0.123	0.630 $\pm$ 0.154	0.670 $\pm$ 0.131	0.650 $\pm$ 0.136	0.7114 $\pm$ 0.108	0.700 $\pm$ 0.098	0.690 $\pm$ 0.095	0.690 $\pm$ 0.099	0.6821 $\pm$ 0.118	0.680 $\pm$ 0.106	0.680 $\pm$ 0.107	0.680 $\pm$ 0.110
EEGConvNeXt [27]	0.7899 $\pm$ 0.058	0.790 $\pm$ 0.054	0.790 $\pm$ 0.056	0.790 $\pm$ 0.056	0.8421 $\pm$ 0.043	0.850 $\pm$ 0.038	0.830 $\pm$ 0.039	0.840 $\pm$ 0.042	0.8001 $\pm$ 0.052	0.800 $\pm$ 0.050	0.800 $\pm$ 0.051	0.800 $\pm$ 0.050
MMSDL [37]	0.7581 $\pm$ 0.071	0.760 $\pm$ 0.074	0.760 $\pm$ 0.071	0.760 $\pm$ 0.072	0.7466 $\pm$ 0.080	0.760 $\pm$ 0.077	0.730 $\pm$ 0.088	0.730 $\pm$ 0.079	0.8091 $\pm$ 0.062	0.810 $\pm$ 0.061	0.810 $\pm$ 0.060	0.810 $\pm$ 0.061
cdRVFL [38]	0.7919 $\pm$ 0.063	0.790 $\pm$ 0.061	0.780 $\pm$ 0.060	0.780 $\pm$ 0.061	0.8142 $\pm$ 0.052	0.810 $\pm$ 0.051	0.810 $\pm$ 0.051	0.810 $\pm$ 0.050	0.7856 $\pm$ 0.057	0.790 $\pm$ 0.058	0.790 $\pm$ 0.057	0.790 $\pm$ 0.058
DCNN-SBiL [39]	0.8268 $\pm$ 0.031	0.830 $\pm$ 0.031	0.830 $\pm$ 0.030	0.830 $\pm$ 0.031	0.8618 $\pm$ 0.021	0.880 $\pm$ 0.020	0.850 $\pm$ 0.019	0.860 $\pm$ 0.021	0.8009 $\pm$ 0.045	0.830 $\pm$ 0.045	0.790 $\pm$ 0.049	0.810 $\pm$ 0.046
SBPFAN [26]	0.8023 $\pm$ 0.033	0.810 $\pm$ 0.032	0.800 $\pm$ 0.032	0.800 $\pm$ 0.031	0.8480 $\pm$ 0.024	0.850 $\pm$ 0.022	0.850 $\pm$ 0.021	0.850 $\pm$ 0.022	0.8229 $\pm$ 0.030	0.820 $\pm$ 0.029	0.840 $\pm$ 0.030	0.820 $\pm$ 0.030
Hjorth-SVM [25]	0.781 $\pm$ 0.028	0.776 $\pm$ 0.031	0.792 $\pm$ 0.029	0.784 $\pm$ 0.027	0.804 $\pm$ 0.025	0.809 $\pm$ 0.026	0.803 $\pm$ 0.025	0.803 $\pm$ 0.025	0.762 $\pm$ 0.030	0.758 $\pm$ 0.031	0.769 $\pm$ 0.029	0.763 $\pm$ 0.028
LDANet	0.8886 $\pm$ 0.018	0.880 $\pm$ 0.017	0.890 $\pm$ 0.018	0.880 $\pm$ 0.018	0.9921 $\pm$ 0.005	0.990 $\pm$ 0.004	0.990 $\pm$ 0.006	0.990 $\pm$ 0.006	0.9057 $\pm$ 0.019	0.920 $\pm$ 0.017	0.910 $\pm$ 0.011	0.910 $\pm$ 0.018

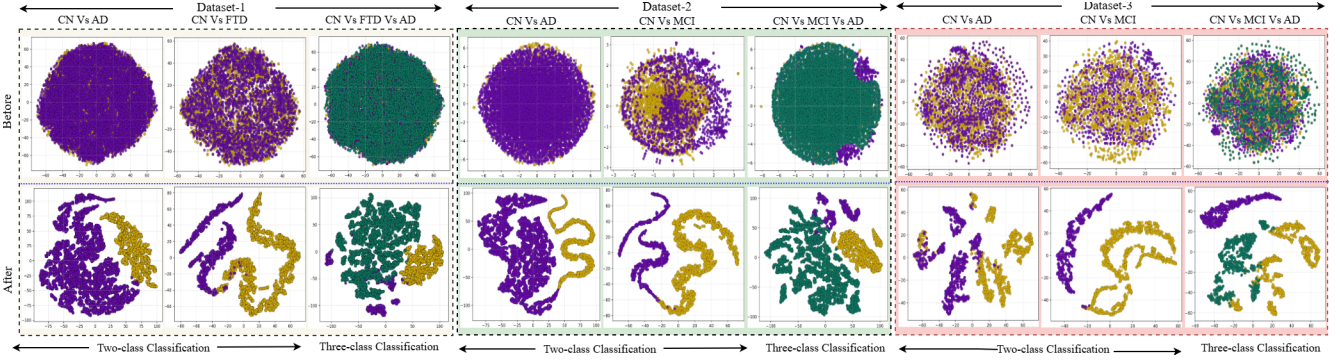


Fig. 5. t-SNE features visualization of the proposed LDANet on datasets-1, -2, and -3.

TABLE VI

A COMPARATIVE ANALYSIS OF THE PROPOSED LDANet WITH SOTA TECHNIQUES IN TERMS OF COMPUTATIONAL COMPLEXITY. HERE, TRAINING TIME IS SHOWN FOR ONE FOLD

Method	#PM	FLOPs	MACs	TTPS	TT (minutes)		
					Dataset-1	Dataset-2	Dataset-3
LCADNet [36]	1.53M	0.009G	0.005G	0.0642	312.6	145.3	42.0
AD-Hjorth [12]				0.0032	8.6	5.2	2.1
VAE-FZ [11]	3.09M	4.46G	2.23G	0.0988	447.3	229.90	87.6
GNN [15]	23.61M	7.60G	3.8G	0.1071	888.4	361.3	180.0
ML-CNN [16]	2.7M	4.46G	2.23G	0.0927	360.11	206.6	66.3
AGGCN [19]	2.28	4.36G	2.18G	0.0682	364.7	197.6	58.3
DEL-AD [21]	37.6M	15.96G	7.98G	0.1566	763.1	264.0	144.2
EEGConvNeXt [27]	7.5M	9.87G	4.94G	0.2311	721.3	310.0	112.6
MMSDL [37]	51.2M	88.6G	44.3G	0.2833	964.3	528.1	285.9
cdRVFL [38]	9.63M	16.2G	8.1G	0.1071	483.2	339.0	189.3
DCNN-SBiL [39]	16.75M	23.8G	11.9G	0.1176	489.6	382.4	126.6
SBPFAN [26]	39.53M	66.2G	33.1G	0.3182	800.32	512.4	165.33
Hjorth-SVM [25]	-	-	-	0.0027	6.3	4.0	1.7
LDANet	2.19M	1.25G	0.62G	0.0613	330.3	182.6	49.0

TABLE VII

WILCOXON RANK-SUM p -VALUES ACROSS THREE DATASETS. HERE, S1 $\rightarrow$ CN Vs. AD, S2 $\rightarrow$ CN Vs. FTD(OR MCI), AND S3 $\rightarrow$ CN Vs. FTD(OR MCI) Vs. AD

Method	Dataset-1			Dataset-2			Dataset-3		
	S1	S2	S3	S1	S2	S3	S1	S2	S3
LCADNet [36]	0.006	0.004	0.012	0.001	0.002	0.004	0.010	0.001	0.007
AD-Hjorth [12]	0.012	0.010	0.018	0.009	0.008	0.015	0.022	0.018	0.027
VAE-FZ [11]	0.001	0.002	0.003	0.001	0.002	0.005	0.006	0.001	0.004
GNN [15]	0.014	0.010	0.021	0.003	0.006	0.010	0.012	0.003	0.016
ML-CNN [16]	0.004	0.006	0.008	0.002	0.004	0.006	0.008	0.002	0.009
AGGCN [19]	0.002	0.002	0.003	0.001	0.001	0.003	0.004	0.001	0.005
DEL-AD [21]	0.003	0.002	0.004	0.001	0.002	0.004	0.005	0.002	0.006
EEGConvNeXt [27]	0.010	0.008	0.015	0.004	0.006	0.012	0.014	0.004	0.018
MMSDL [37]	0.009	0.006	0.011	0.003	0.007	0.009	0.011	0.006	0.008
cdRVFL [38]	0.025	0.019	0.017	0.007	0.011	0.015	0.016	0.005	0.021
DCNN-SBiL [39]	0.018	0.022	0.028	0.006	0.010	0.013	0.024	0.010	0.033
SBPFAN [26]	0.033	0.028	0.041	0.021	0.017	0.061	0.029	0.013	0.041
Hjorth-SVM [25]	0.018	0.015	0.024	0.013	0.011	0.019	0.029	0.024	0.034

the results, LDANet consistently outperforms all competing methods across all three datasets. This superior performance is attributed to four main factors: (1) the development of an efficient end-to-end deep learning framework that captures

both HL and LL features; (2) the inclusion of an RDSPP module that extracts features at multiple levels; (3) the integration of an attention mechanism that highlights relevant contextual information; and (4) the use of a constraint-based fusion approach that effectively merges HL and LL characteristics. Among the existing methods, SBPFAN [26] achieves the second-best results, showing comparable performance to LDANet on Dataset-1 and Dataset-2. However, it falls behind on Dataset-3, where LDANet shows clear advantages. In addition to accuracy-based comparisons, LDANet is also evaluated against SOTA methods in terms of computational complexity. Table VI summarizes metrics such as the number of parameters (#PM), floating-point operations (FLOPs), multiply-accumulate operations (MACs), training time per fold (TT), and testing time per sample (TTPS), across all three datasets. The experiments were conducted on an NVIDIA Tesla T4 GPU (16 GB memory), and the reported times were shown for one fold. LDANet required 0.06 milliseconds per test sample, while maintaining a total memory footprint below 12.1 GB of GPU during training. The training and testing durations vary depending on factors like model size, input resolution, number of epochs, and fold count. As shown in Table VI, LDANet achieved better results while maintaining a lower computational load than other competing methods, making it not only accurate but also efficient.

Further, the feature visualization of the proposed LDANet is evaluated using t-SNE plot. The t-SNE method facilitates the visualization of intricate data by transforming high-dimensional datasets into two- or three-dimensional representations. The t-SNE is utilized to visualize the data representations before and after the model processes them.

TABLE VIII
COMPARATIVE ANALYSIS OF THE PROPOSED LDANet ON DATASETS-1, -2 AND -3 FOR A SUBJECT-INDEPENDENT EVALUATION

Method	Dataset-1			Dataset-2			Dataset-3		
	CN vs. AD	CN vs. FTD	CN vs. FTD vs. AD	CN vs. AD	CN vs. MCI	CN vs. MCI vs. AD	CN vs. AD	CN vs. MCI	CN vs. MCI vs. AD
LCADNet [36]	0.8336 ± 0.052	0.8511 ± 0.048	0.8195 ± 0.072	0.8327 ± 0.059	0.8001 ± 0.074	0.7916 ± 0.069	0.6192 ± 0.091	0.6582 ± 0.083	0.6474 ± 0.095
AD-Hjorth [12]	0.8428 ± 0.037	0.8264 ± 0.041	0.8017 ± 0.049	0.8593 ± 0.034	0.8336 ± 0.039	0.8129 ± 0.046	0.7521 ± 0.056	0.7735 ± 0.052	0.7398 ± 0.059
VAE-FZ [11]	0.7963 ± 0.067	0.8321 ± 0.055	0.7855 ± 0.079	0.7718 ± 0.065	0.8308 ± 0.049	0.7716 ± 0.089	0.5992 ± 0.092	0.6202 ± 0.086	0.6144 ± 0.094
GNN [15]	0.8532 ± 0.049	0.8201 ± 0.053	0.7918 ± 0.068	0.8452 ± 0.042	0.8105 ± 0.076	0.8417 ± 0.062	0.6890 ± 0.074	0.6418 ± 0.082	0.6315 ± 0.089
ML-CNN [16]	0.8563 ± 0.045	0.8712 ± 0.036	0.8059 ± 0.082	0.8336 ± 0.063	0.7919 ± 0.074	0.7081 ± 0.095	0.6993 ± 0.076	0.7537 ± 0.057	0.6874 ± 0.084
AGGCN [19]	0.8219 ± 0.058	0.8472 ± 0.051	0.8609 ± 0.043	0.8714 ± 0.029	0.7629 ± 0.083	0.7016 ± 0.091	0.7688 ± 0.061	0.7004 ± 0.072	0.6993 ± 0.076
DEL-AD [21]	0.8336 ± 0.061	0.8001 ± 0.068	0.7891 ± 0.071	0.7525 ± 0.087	0.6892 ± 0.093	0.7096 ± 0.096	0.6584 ± 0.078	0.6928 ± 0.084	0.7024 ± 0.092
EEGConvNeXt [27]	0.8319 ± 0.057	0.8201 ± 0.065	0.7866 ± 0.079	0.7027 ± 0.092	0.7419 ± 0.083	0.7011 ± 0.098	0.7542 ± 0.063	0.7128 ± 0.072	0.6917 ± 0.080
MMSDL [37]	0.8681 ± 0.039	0.8930 ± 0.034	0.8411 ± 0.051	0.7638 ± 0.072	0.7869 ± 0.063	0.8093 ± 0.058	0.7516 ± 0.061	0.8001 ± 0.059	0.7326 ± 0.067
edRVFL [38]	0.9133 ± 0.025	0.9352 ± 0.019	0.8001 ± 0.062	0.8093 ± 0.051	0.8419 ± 0.044	0.8792 ± 0.047	0.7293 ± 0.069	0.7937 ± 0.054	0.7090 ± 0.074
DCNN-SBIL [39]	0.9016 ± 0.018	0.9115 ± 0.021	0.8419 ± 0.049	0.9066 ± 0.022	0.8917 ± 0.023	0.8552 ± 0.051	0.7916 ± 0.058	0.7846 ± 0.049	0.7933 ± 0.066
SBPFAN [26]	0.9474 ± 0.013	0.9018 ± 0.016	0.8856 ± 0.027	0.9701 ± 0.009	0.9245 ± 0.017	0.8708 ± 0.039	0.8011 ± 0.052	0.8164 ± 0.045	0.7917 ± 0.061
Hjorth-SVM [25]	0.8123 ± 0.041	0.7987 ± 0.046	0.7814 ± 0.053	0.8269 ± 0.038	0.8011 ± 0.045	0.7832 ± 0.049	0.7326 ± 0.056	0.7484 ± 0.052	0.7159 ± 0.061
LDANet	0.9626 ± 0.007	0.9698 ± 0.006	0.9368 ± 0.011	0.9711 ± 0.005	0.9518 ± 0.008	0.9248 ± 0.013	0.8503 ± 0.045	0.9311 ± 0.018	0.8581 ± 0.027

These visualizations illustrate how LDANet modifies the data to enhance the distinguishability of various classes. Figure 5 illustrates the t-SNE charts for three datasets. Owing to memory constraints and the substantial size of the training data in Dataset-1, few data points are randomly selected for t-SNE chart.

Comparing the plots before and after the application of LDANet elucidates the rationale behind its categorization decisions.

Moreover, to assess the effectiveness of the proposed method, we tested whether LDANet's gains are reliable using the Wilcoxon rank-sum test, a non-parametric test for independent samples [26]. All tests were run at the 5 % level. For each of the 13 baseline methods and for LDANet, we computed ten scores via 10-fold cross-validation on three datasets, giving ten independent values per method-dataset pair. Let E_j be the median performance of method j ($j=1$ denotes LDANet; $j=2, \dots, 13$ denote baselines). For each baseline $l \in \{2, \dots, 13\}$ we evaluated $H_0 : E_1 = E_l$ vs. $H_a : E_1 > E_l$, i.e., a one-sided superiority test favoring LDANet. Folds are disjoint across methods, so samples are treated as unpaired, matching the rank-sum assumptions. Table VII lists the resulting p -values for the three datasets. The majority fall below 0.05, providing evidence to reject H_0 and conclude that LDANet's median scores are statistically higher than those of most competing methods. The smallest p -values are observed for Dataset-2 and Dataset-3 in the binary classification tasks, reflecting the large performance margins achieved by LDANet. Only a single comparison (Dataset-2, CN vs. MCI vs. AD) against SBPFAN did not reach statistical significance, which can be attributed to the closely matched accuracy scores between the two models. Overall, these results provide strong statistical evidence that LDANet consistently and significantly outperforms existing deep learning architectures across diverse EEG-based dementia classification scenarios.

5) *Subject-Independent Evaluation*: In this section, subject-independent evaluation is considered to find the efficacy of the proposed LDANet for Dataset-1, Dataset-2, and Dataset-3. To ensure class balance, folds were generated using a class-stratified subject grouping strategy. In Dataset-1, 88 subjects are divided into ten distinct groups, ensuring no subject is included in more than one group. Out of these ten groups, eight comprise data from 8 subjects each, while the remaining two consist of data from 12 subjects each. To achieve a balanced

TABLE IX
COMPARATIVE ANALYSIS OF CROSS-DATASET EVALUATION OF THE PROPOSED LDANet

Method	Dataset-1				Dataset-2			
	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$	$\mathcal{A}$	$\mathcal{P}$	$\mathcal{R}$	$\mathcal{F}$
LCADNet [36]	0.7942	0.78	0.77	0.78	0.8251	0.81	0.80	0.81
AD-Hjorth [12]	0.7416	0.72	0.71	0.72	0.7734	0.75	0.74	0.75
VAE-FZ [11]	0.7835	0.76	0.75	0.75	0.8123	0.79	0.78	0.79
GNN [15]	0.7764	0.75	0.74	0.74	0.8056	0.78	0.77	0.77
ML-CNN [16]	0.7689	0.74	0.73	0.73	0.7984	0.76	0.76	0.76
AGGCN [19]	0.8025	0.79	0.78	0.78	0.8338	0.81	0.80	0.80
DEL-AD [21]	0.7846	0.76	0.75	0.75	0.8162	0.79	0.78	0.78
EEGConvNeXt [27]	0.8143	0.80	0.79	0.79	0.8452	0.83	0.82	0.82
MMSDL [37]	0.7996	0.78	0.77	0.77	0.8297	0.80	0.79	0.79
edRVFL [38]	0.7624	0.73	0.72	0.72	0.7901	0.75	0.74	0.74
DCNN-SBIL [39]	0.7753	0.75	0.74	0.74	0.8035	0.78	0.77	0.77
SBPFAN [26]	0.8357	0.82	0.81	0.83	0.8829	0.87	0.86	0.87
Hjorth-SVM [25]	0.7123	0.70	0.68	0.69	0.7398	0.72	0.71	0.72
LDANet	0.8618	0.85	0.84	0.86	0.9203	0.91	0.90	0.91

TABLE X
THE PERFORMANCE ANALYSIS OF THE LDANet WITH DIFFERENT WINDOW SIZES AND OVERLAPPING RATES. HERE, S1 → CN Vs. AD, S2 → CN Vs. FTD/MCI, AND S3 → CN Vs. FTD/MCI Vs. AD

Window size	Varying window size								
	Dataset-1			Dataset-2			Dataset-3		
	S1	S2	S3	S1	S2	S3	S1	S2	S3
2 sec	0.9654	0.9688	0.9732	0.9891	0.9823	0.9710	0.8710	0.9695	0.8940
4 sec	0.9890	0.9905	0.9963	0.9998	0.9982	0.9835	0.8886	0.9921	0.9057
6 sec	0.9820	0.9841	0.9890	0.9957	0.9915	0.9750	0.8805	0.9810	0.9020
8 sec	0.9702	0.9735	0.9788	0.9920	0.9860	0.9632	0.8728	0.9733	0.9000
No overlap	Varied overlapping window rates								
	25%	0.9690	0.9708	0.9760	0.9910	0.9848	0.9620	0.8705	0.9711
	50%	0.9832	0.9855	0.9912	0.9973	0.9935	0.9794	0.8841	0.9857
	75%	0.9890	0.9905	0.9963	0.9998	0.9835	0.8886	0.9921	0.9057
	No overlap	0.9628	0.9663	0.9715	0.9876	0.9805	0.9593	0.8650	0.9662

distribution, each group includes at least one subject from CN, FTD (or MCI), and AD. This ensures a uniform distribution of subjects across the groups. Similarly in Dataset-2, 168 subjects are divided into ten distinct groups. Out of these ten groups, eight comprise data from 17 subjects each, while the remaining two consist of data from 16 subjects each. In Dataset-2, only 7 subjects are available for MCI, so only 7 groups have 1 MCI subject each. Similarly, Dataset-3 is divided into 10 groups. Out of these ten groups, five comprise data from 4 subjects each, while the remaining five consist of data from 3 subjects each. In Dataset-3, only 7 subjects are available for MCI, so only 7 groups have 1 MCI subject each. LDANet is then trained using ten-fold cross-validation, where one group serves as the test set in each fold, and the other nine groups are used for training and validation. Data from the nine groups are shuffled randomly before being split into an 80% training set and a 20% validation set. The comparative experimental results of Dataset-1, Dataset-2, and Dataset-3 are reported in Table VIII. Ten-fold cross-validation was applied uniformly

TABLE XI

THE INVESTIGATION OF VARIOUS MODULES IN THE PROPOSED LDANet IS PERFORMED USING DATASET-1, -2, AND -3. ADDITIONALLY, WITHIN THE FRAMEWORK, DF STANDS FOR DIRECT FUSION

LL	HL	RDSPP	SA	CA	CFM	DF	Dataset-1			Dataset-2			Dataset-3		
							CN vs. AD	CN vs. FTD	CN vs. AD	CN vs. AD	CN vs. MCI	CN vs. AD	CN vs. AD	CN vs. MCI	CN vs. MCI vs. AD
✓	✗	✗	✗	✗	✗	✗	0.7506	0.7602	0.6814	0.7813	0.7741	0.6523	0.7261	0.7480	0.6672
✓	✓	✗	✗	✗	✗	✓	0.8721	0.7785	0.8894	0.8910	0.8842	0.8673	0.7505	0.8633	0.7799
✓	✓	✗	✗	✗	✓	✗	0.8860	0.8803	0.9011	0.9142	0.9184	0.9012	0.8190	0.9201	0.8252
✓	✗	✓	✗	✗	✗	✗	0.8547	0.8629	0.7836	0.7856	0.7792	0.8568	0.7538	0.8532	0.8001
✓	✗	✓	✓	✗	✗	✗	0.8611	0.8885	0.8462	0.8593	0.8821	0.8615	0.8012	0.8786	0.8350
✓	✗	✓	✗	✗	✗	✗	0.8535	0.8714	0.7975	0.8402	0.8833	0.8240	0.7937	0.8614	0.7776
✗	✓	✗	✗	✗	✗	✗	0.7880	0.7578	0.7096	0.7982	0.7700	0.6995	0.7889	0.7440	0.6924
✗	✓	✓	✗	✗	✗	✗	0.8039	0.8210	0.8024	0.8317	0.8346	0.7840	0.7974	0.8095	0.8087
✗	✓	✓	✓	✗	✗	✗	0.8992	0.8658	0.8849	0.8865	0.8780	0.8586	0.8549	0.8551	0.8625
✗	✓	✓	✗	✓	✗	✗	0.9418	0.9582	0.9661	0.9579	0.9601	0.9409	0.8483	0.9380	0.8843
✓	✓	✓	✓	✓	✗	✓	0.9792	0.9834	0.9830	0.9766	0.9730	0.9680	0.8512	0.9587	0.8823
✓	✓	✓	✓	✓	✓	✗	0.9890	0.9905	0.9963	0.9998	0.9982	0.9835	0.8886	0.9921	0.9057

across all 13 methods listed in Table VIII using the same datasets. For consistency, each fold utilized identical training, validation, and testing sets across all methods, including LDANet. It is observed from Table VIII that the subject-independent evaluation shows almost similar accuracy in both datasets compared to the results of Table III, IV and V.

6) *Cross-Dataset Evaluation:* The goal of the cross-dataset evaluation is to test the performance of LDANet on data it hasn't seen during training. This helps us to understand how adaptable the model is when faced with new data that may have different noise levels, data distributions, or other variations. In this evaluation, we trained LDANet on Dataset-1 and tested it on Dataset-2, and then did the reverse training on Dataset-2 and testing on Dataset-1. This allowed us to confirm that LDANet maintains strong performance across different but related datasets, especially since both datasets involve similar AD vs. CN classification tasks. We excluded Dataset-3 from this evaluation because its number of channels did not match those in Datasets 1 and 2, making a fair comparison impossible. The results shown in Table IX, demonstrate that LDANet consistently outperforms current SOTA methods on the test datasets. In fact, its accuracy in cross-dataset is almost similar to both subject-dependent and subject-independent evaluation. This highlights LDANet's strong generalization ability and suggests it could be highly effective in practical, real-world applications.

D. Ablation Study

In this study, we propose several enhancements that, when combined, contribute to the improved performance of our novel method across the three datasets. To better understand the individual contribution of each component, we conduct an ablation study. This involves systematically disabling specific components and observing the resulting drop in performance, thereby quantifying their individual impact. As part of the preprocessing stage, raw EEG signals are divided into 4-second segments with a 75% overlap. The rationale for this segmentation approach is detailed in Section III-B. To further examine the effect of different segmentation configurations, we perform an experiment using various window durations and overlapping rates. The results of this evaluation are summarized in Table X. As shown, the 4-second window configuration consistently achieves the highest performance. Likewise, a 75% overlap yields superior results compared to other overlap settings, although the differences are relatively

minor, indicating that the method remains robust across a range of overlapping rates.

Further, the processed data is given to LDANet for training and evaluation. LDANet consists of six primary components that contribute to its superior performance: LL and HL features, RDSPP, SA, CA, and CFM. To assess the contribution of each module, we perform extensive ablation studies by enabling or disabling individual components and evaluating performance across three datasets. The results of these experiments are reported in Table XI, where each row presents a unique configuration of the LDANet components. A ✓ symbol indicates that a specific module is included, while a ✗ symbol indicates it is excluded. The results clearly show that the complete model (last row), which includes all six modules, achieves the highest accuracy across all binary and multiclass classification tasks. Importantly, the use of the CFM yields consistent improvements compared to Direct Fusion (DF). This demonstrates that CFM effectively consolidates the contributions of HL and LL features by adaptively assigning weights to each, based on their contextual relevance. Furthermore, the inclusion of RDSPP, SA, and CA modules also shows significant improvements, particularly in multiclass classification tasks. Removing any of these modules causes performance degradation, underscoring their importance.

V. CONCLUSION

This study introduces a novel architecture LDANet, designed for effective detection of AD. Central to LDANet is the RDSPP module, which processes both HL and LL features to extract the most salient and distinctive representations. These selected features are subsequently passed to a multi-level attention mechanism, which further emphasizes the most diagnostically relevant characteristics. LDANet also incorporates a fully connected, weight-sharing layer, which strengthens the correlation between HL and LL feature learning, enhancing overall representational quality. Furthermore, a CFM adaptively assigns fusion weights based on the relative importance of HL and LL features, enabling the network to effectively leverage complementary information across different feature hierarchies. This adaptive fusion contributes significantly to the model's capacity to accurately distinguish AD-related patterns. The performance of LDANet is rigorously evaluated on three benchmark datasets, demonstrating consistent improvements over existing methods. Comprehensive ablation studies further validate the

contribution of individual module to the overall performance. Notably, LDANet surpasses 13 SOTA methods in AD classification accuracy, underlining its potential for clinical application. Despite its high efficacy, LDANet maintains a low computational footprint, comprising only 2.19 million parameters and a model size of 9.87 MB, which supports its deployment in real-time and resource-constrained environments.

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